

SSILP Math

Beginner

DIVIDING POLYNOMIALS (SELF STUDY)



Learning objectives

- **Understand and Perform Long Division of Polynomials**
- **Apply the Factor Theorem to factor polynomials**

Session outline

1. Dividing polynomials

1. Dividing polynomials

Long division of polynomials

Diagram illustrating the components of the division $\frac{38}{7} = 5 + \frac{3}{7}$:

- Dividend:** 38
- Divisor:** 7
- Quotient:** 5
- Remainder:** 3

Dividing polynomials is much like the familiar process of dividing numbers

To divide polynomials, we use **long division**

DIVISION ALGORITHM

If $P(x)$ and $D(x)$ are polynomials, with $D(x) \neq 0$, then there exist unique polynomials $Q(x)$ and $R(x)$, where $R(x)$ is either 0 or of degree less than the degree of $D(x)$, such that

$$\frac{P(x)}{D(x)} = Q(x) + \frac{R(x)}{D(x)} \quad \text{or} \quad P(x) = D(x) \cdot Q(x) + R(x)$$

Diagram illustrating the components of the polynomial division equation $P(x) = D(x) \cdot Q(x) + R(x)$:

- Dividend:** $P(x)$
- Divisor:** $D(x)$
- Quotient:** $Q(x)$
- Remainder:** $R(x)$

The polynomials $P(x)$ and $D(x)$ are called the **dividend** and **divisor**, respectively, $Q(x)$ is the **quotient**, and $R(x)$ is the **remainder**.

Long division of polynomials

Divide $6x^2 - 26x + 12$ by $x - 4$

EXAMPLE

Solution: Long division of polynomials

SOLUTION The *dividend* is $6x^2 - 26x + 12$, and the *divisor* is $x - 4$. We begin by arranging them as follows.

$$x - 4 \overline{) 6x^2 - 26x + 12}$$

Next we divide the leading term in the dividend by the leading term in the divisor to get the first term of the quotient: $6x^2/x = 6x$. Then we multiply the divisor by $6x$ and subtract the result from the dividend.

$$\begin{array}{r} \overline{6x} \\ x - 4 \overline{) 6x^2 - 26x + 12} \\ \underline{6x^2 - 24x} \\ -2x + 12 \end{array}$$

Divide leading terms: $\frac{6x^2}{x} = 6x$

Multiply: $6x(x - 4) = 6x^2 - 24x$

Subtract and “bring down” 12

We repeat the process using the last line $-2x + 12$ as the dividend.

$$\begin{array}{r} \overline{6x - 2} \\ x - 4 \overline{) 6x^2 - 26x + 12} \\ \underline{6x^2 - 24x} \\ -2x + 12 \\ \underline{-2x + 8} \\ 4 \end{array}$$

Divide leading terms: $\frac{-2x}{x} = -2$

Multiply: $-2(x - 4) = -2x + 8$

Subtract

EXAMPLE

Solution: Long division of polynomials

The division process ends when the last line is of lesser degree than the divisor. The last line then contains the *remainder*, and the top line contains the *quotient*. The result of the division can be interpreted in either of two ways:

$$\begin{array}{c} \text{Dividend} \\ \hline 6x^2 - 26x + 12 \\ \hline \text{Divisor} \quad x - 4 \end{array} = \begin{array}{c} \text{Quotient} \\ 6x - 2 \end{array} + \begin{array}{c} \text{Remainder} \\ \frac{4}{x - 4} \end{array} \quad \text{or} \quad 6x^2 - 26x + 12 = \begin{array}{c} \text{Dividend} \\ (x - 4) \end{array} \begin{array}{c} \text{Divisor} \\ (6x - 2) \end{array} + \begin{array}{c} \text{Remainder} \\ 4 \end{array}$$

The diagram illustrates the long division of the polynomial $6x^2 - 26x + 12$ by the linear divisor $x - 4$. The result is shown in two equivalent forms. In the first form, the dividend is divided by the divisor to yield a quotient of $6x - 2$ and a remainder of 4 , expressed as a fraction $\frac{4}{x - 4}$. In the second form, the same division is expressed as an equation: the dividend equals the product of the divisor and the quotient, plus the remainder. Labels with callout boxes identify the components: Dividend, Divisor, Quotient, and Remainder.

EXAMPLE

Example: Long division of polynomials

Let $P(x) = 8x^4 + 6x^2 - 3x + 1$ and $D(x) = 2x^2 - x + 2$. Find polynomials $Q(x)$ and $R(x)$ such that $P(x) = D(x) \cdot Q(x) + R(x)$.

EXAMPLE

Long division of polynomials

Let $P(x) = 8x^4 + 6x^2 - 3x + 1$ and $D(x) = 2x^2 - x + 2$. Find polynomials $Q(x)$ and $R(x)$ such that $P(x) = D(x) \cdot Q(x) + R(x)$.

SOLUTION We use long division after first inserting the term $0x^3$ into the dividend to ensure that the columns line up correctly.

$$\begin{array}{r}
 \overline{8x^4 + 0x^3 + 6x^2 - 3x + 1} \\
 \underline{8x^4 - 4x^3 + 8x^2} \\
 4x^3 - 2x^2 - 3x + 1 \\
 \underline{4x^3 - 2x^2 + 4x} \\
 -7x + 1
 \end{array}$$

$4x^2 + 2x$
 Multiply divisor by $4x^2$
 Subtract
 Multiply divisor by $2x$
 Subtract

The process is complete at this point because $-7x + 1$ is of lesser degree than the divisor $2x^2 - x + 2$. From the above long division we see that $Q(x) = 4x^2 + 2x$ and $R(x) = -7x + 1$, so

$$8x^4 + 6x^2 - 3x + 1 = (2x^2 - x + 2)(4x^2 + 2x) + (-7x + 1)$$

EXAMPLE

Factor Theorem

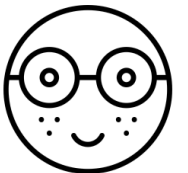
FACTOR THEOREM

c is a zero of P if and only if $x - c$ is a factor of $P(x)$.

“If and only if” means
that the converse is also
true!

We use this theorem when factoring the polynomials.

It links **factors** and **zeros** of the polynomial.



Factoring a polynomial using the factor theorem

Let $P(x) = x^3 - 7x + 6$. Show that $P(1) = 0$, and use this fact to factor $P(x)$ completely.

SOLUTION Substituting, we see that $P(1) = 1^3 - 7 \cdot 1 + 6 = 0$. By the Factor Theorem this means that $x - 1$ is a factor of $P(x)$. Using synthetic or long division (shown in the margin), we see that

$$P(x) = x^3 - 7x + 6 \quad \text{Given polynomial}$$

$$= (x - 1)(x^2 + x - 6)$$

$$= (x - 1)(x - 2)(x + 3) \quad \text{Factor quadratic } x^2 + x - 6$$

EXAMPLE

Two polynomials P and D are given. Use long division to divide $P(x)$ by $D(x)$

(a) $P(x) = 2x^2 - 5x - 7, \quad D(x) = x - 2$

(b) $P(x) = 3x^3 + 9x^2 - 5x - 1, \quad D(x) = x + 4$

(c) $P(x) = 6x^3 + x^2 - 12x + 5, \quad D(x) = 3x - 4$

2.1 Dividing polynomials

(a) $P(x) = 2x^2 - 5x - 7$, $D(x) = x - 2$

$$\begin{array}{r}
 2x - 1 \\
 x - 2 \overline{) 2x^2 - 5x - 7} \\
 \underline{2x^2 - 4x} \quad \text{Subtract} \\
 -x - 7 \\
 \underline{-x + 2} \quad \text{Subtract} \\
 -9
 \end{array}$$

$$\frac{2x^2 - 5x - 7}{x - 2} = 2x - 1 + \frac{-9}{x-2}$$

2.1 Dividing polynomials

(b) $P(x) = 3x^3 + 9x^2 - 5x - 1$, $D(x) = x + 4$

$$\begin{array}{r}
 3x^2 - 3x + 7 \\
 x + 4 \overline{) 3x^3 + 9x^2 - 5x - 1} \\
 \underline{3x^3 + 12x^2} \quad \text{Subtract} \\
 -3x^2 - 5x - 1 \\
 \underline{-3x^2 - 12x} \quad \text{Subtract} \\
 7x - 1 \\
 \underline{7x + 28} \quad \text{Subtract} \\
 -29
 \end{array}$$

$$\frac{3x^3 + 9x^2 - 5x - 1}{x + 4} = 3x^2 - 3x + 7 + \frac{-29}{x+4}$$

2.1 Dividing polynomials

(c) $P(x) = 6x^3 + x^2 - 12x + 5$, $D(x) = 3x - 4$

$$\begin{array}{r}
 2x^2 + 3x \\
 3x - 4 \overline{) 6x^3 + x^2 - 12x + 5} \\
 \underline{6x^3 - 8x^2} \qquad \text{Subtract} \\
 9x^2 - 12x + 5 \\
 \underline{9x^2 - 12x} \qquad \text{Subtract} \\
 5
 \end{array}$$

$$\frac{6x^3 + x^2 - 12x + 5}{3x - 4} = 2x^2 + 3x + \frac{5}{3x-4}$$

Finding a polynomial with specified zeros

Find a polynomial of the specified degree that has the given zeros.

(a) Degree 3; zeros $-1, 1, 3$

(b) Degree 4; zeros $-2, 0, 2, 4$

ACTIVITY 1.2

2.2 Finding a polynomial with specified zeros

Degree 3; zeros $-1, 1, 3$

If c is a zero of a polynomial, then $x - c$ is a factor of the polynomial.

Since $-1, 1$, and 3 are zeros of the polynomial, then $(x + 1)$, $(x - 1)$, and $(x - 3)$ are factors of the polynomial.

Thus, one polynomial of degree 3 with the given zeros is:

$$P(x) = (x + 1)(x - 1)(x - 3)$$

$$P(x) = x^3 - 3x^2 - x + 3$$

2.2 Finding a polynomial with specified zeros

Degree 4; zeros $-2, 0, 2, 4$

If c is a zero of a polynomial, then $x - c$ is a factor of the polynomial.

Since $-2, 0, 2$, and 4 are zeros of the polynomial, then $(x + 2)$, (x) , $(x - 2)$, and $(x - 4)$ are factors of the polynomial.

Thus, one polynomial of degree 4 with the given zeros is:

$$P(x) = (x + 2)(x)(x - 2)(x - 4)$$

$$P(x) = x^4 - 4x^3 - 4x^2 + 16x$$